Lecture 13

The Gamma function and Laplace's method

Objectives. *Euler's integral defines the Gamma function in a half-plane. Its functional equation extends the function meromorphically to the plane. We use the integral to evaluate radial integrals and to prove real Stirling asymptotics. A cited sectorial estimate then controls the logarithmic derivative.*

By the end of this lecture you will be able to:

- Continue Euler's integral and compute its poles and residues.
- Evaluate Beta and convergent radial integrals with their parameter conditions.
- Prove the leading Laplace approximation at a unique interior minimum.
- Derive a digamma estimate from the cited sectorial Stirling estimate.

13.1 Euler's integral and meromorphic continuation

The *Gamma function* is initially defined by

$$\Gamma(s) = \int_0^\infty t^{s-1} e^{-t}\, \mathrm{d}t, \qquad \operatorname{Re} s > 0, \qquad t^{s-1} = e^{(s-1)\ln t}. \tag{13.1}$$

The symbol $\Gamma(s)$ always has an argument. It is different from the decay rate Γ used without an argument in Lecture 12. The powers in this integral use a positive real base.

For a compact parameter set in the right half-plane choose $0 < c \le \operatorname{Re} s \le C$. On $(0,1)$ the modulus of the integrand is at most t^{c-1}. On $[1,\infty)$ it is at most $t^{C-1}e^{-t}$. Both bounds are integrable. The holomorphic parameter theorem, Theorem 10.9, therefore makes the Gamma function holomorphic in this half-plane and permits its complex derivatives under the integral.

Integration by parts gives the functional equation

$$\Gamma(s+1) = s\Gamma(s), \qquad \operatorname{Re} s > 0. \tag{13.2}$$

The boundary term $t^s e^{-t}$ tends to zero at both endpoints: its modulus is $t^{\operatorname{Re} s} e^{-t}$. Direct integration gives $\Gamma(1) = 1$, so induction gives $\Gamma(n+1) = n!$ for integers $n \ge 0$.

Theorem 13.1. *The Gamma function has a meromorphic continuation to $\mathbb{C}$. Its only poles are simple poles at $s = -n$, for integers $n \ge 0$, with*

$$\operatorname{Res}_{s=-n} \Gamma(s) = \frac{(-1)^n}{n!}. \tag{13.3}$$

The functional equation holds as a meromorphic identity.

Proof. For each integer $N \geq 1$, define the extension in $\operatorname{Re} s > -N$ by

$$\Gamma(s) = \frac{\Gamma(s+N)}{s(s+1)\cdots(s+N-1)}.$$

The numerator uses Euler's integral, since $\operatorname{Re}(s+N) > 0$. Successive definitions agree wherever both denominators are nonzero by the functional equation. Their meromorphic extensions therefore agree also at removable points. These half-planes cover $\mathbb{C}$.

At $s = -n$, use $N = n+1$. The numerator is $\Gamma(1) = 1$. After removing the factor $s+n$ from the denominator, its value is $(-n)(1-n)\cdots(-1) = (-1)^n n!$, with empty product 1 when $n = 0$. Thus the pole is simple and has the stated residue. There are no other possible poles. The functional equation persists on every half-plane by the same defining quotients. ∎

For example, the extension assigns a finite value at $s = -1/2$ even though Euler's integral diverges there. The functional equation gives $\Gamma(-1/2) = -2\Gamma(1/2)$. The original divergent integral has not acquired a value as an improper integral.

13.2 Beta and reflection

For $\operatorname{Re} a > 0$ and $\operatorname{Re} b > 0$, the *Beta function* is

$$B(a,b) = \int_0^1 t^{a-1}(1-t)^{b-1}\,\mathrm{d}t.$$

Both endpoint powers are absolutely integrable under these conditions.

Theorem 13.2. *For* $\operatorname{Re} a > 0$ *and* $\operatorname{Re} b > 0$,

$$B(a,b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}. \tag{13.4}$$

The Gamma function has no zeros. For $s \in \mathbb{C} \setminus \mathbb{Z}$,

$$\Gamma(s)\Gamma(1-s) = \frac{\pi}{\sin \pi s}. \tag{13.5}$$

In particular, $\Gamma(1/2) = \sqrt{\pi}$.

Proof. The product of the two Euler integrals is absolutely integrable on $(0,\infty)^2$. Fubini and change of variables, standing facts M4 and M5, Theorems 4.1 and 4.2, allow $x = ut$ and $y = u(1-t)$, where $u > 0$, $0 < t < 1$ and the absolute Jacobian is u. Thus

$$\Gamma(a)\Gamma(b) = \int_0^\infty \int_0^1 e^{-u} u^{a+b-1} t^{a-1}(1-t)^{b-1}\,\mathrm{d}t\,\mathrm{d}u = \Gamma(a+b)B(a,b).$$

For now retain this product identity without dividing.

Set $b = 1-a$ with $0 < a < 1$. The substitution $t = x/(1+x)$ gives

$$B(a,1-a) = \int_0^\infty \frac{x^{a-1}}{1+x}\,\mathrm{d}x = \frac{\pi}{\sin \pi a}$$

by the proved keyhole formula, Equation (11.4). Since $\Gamma(1) = 1$, reflection follows first on this real interval. Both sides are holomorphic on the connected open set $\mathbb{C} \setminus \mathbb{Z}$. The identity theorem, Theorem 10.6, extends the equality there. This set is connected because any polygonal path can be detoured around the finitely many integers it meets.

The right side of reflection never vanishes. Hence $\Gamma(s)$ cannot vanish at any noninteger. At positive integers it equals a nonzero factorial. The remaining integers are poles. This proves the absence of zeros and permits division by $\Gamma(a+b)$ in the product identity. Finally reflection at $s = 1/2$ gives $\Gamma(1/2)^2 = \pi$. Euler's positive integrand selects the positive square root. ∎

For example, $B(2,3) = 1!2!/4! = 1/12$. This agrees with direct integration of $t(1-t)^2$ on $[0,1]$. The absence of zeros also makes $1/\Gamma(s)$ entire as a function of s: it extends by zero at each simple pole $s = -n$ of $\Gamma(s)$.

13.3 Sphere areas and radial integrals

For a positive integer d, the symbol $\left|S^{d-1}\right|$ means the surface area of the unit sphere in $\mathbb{R}^d$. This is its $(d-1)$-dimensional measure. Its d-dimensional Lebesgue measure is zero. The bars here denote surface area, rather than volume or modulus.

Tonelli's theorem, Theorem 4.1, separates the product of nonnegative one-dimensional Gaussians. With Theorem 4.3 this gives $\int_{\mathbb{R}^d} e^{-|k|^2}\,\mathrm{d}^d k = \pi^{d/2}$. The radial change-of-variables formula M5, Theorem 4.2, gives the same integral as $\left|S^{d-1}\right| \int_0^\infty r^{d-1} e^{-r^2}\,\mathrm{d}r$. Substituting $t = r^2$ evaluates this last integral as $\Gamma(d/2)/2$. Consequently

$$\left|S^{d-1}\right| = \frac{2\pi^{d/2}}{\Gamma(d/2)}. \tag{13.6}$$

For $d = 1$ this is 2, counting the two points of the unit sphere. For $d = 3$ it is 4π.

Let $m > 0$ and $\operatorname{Re} s > d/2$. The following integral converges absolutely: near infinity its radial modulus is $O(r^{d-1-2\operatorname{Re} s})$, and near zero it is $O(r^{d-1})$. Substituting $u = r^2/m^2$ gives

$$\begin{aligned}
&\int_{\mathbb{R}^d} (|k|^2 + m^2)^{-s}\,\mathrm{d}^d k \\
&\quad = \frac{\left|S^{d-1}\right|}{2} m^{d-2s} \\
&\quad \times \int_0^\infty u^{d/2-1}(1+u)^{-s}\,\mathrm{d}u \\
&\quad = \frac{\pi^{d/2}\Gamma(s-d/2)}{\Gamma(s)} m^{d-2s}.
\end{aligned}$$

The last integral is $B(d/2, s-d/2)$ by $t = u/(1+u)$. Every power of a positive quantity uses its real logarithm. For example, $d = 3$, $s = 2$ and $m = 2$ give the value $\pi^2/2$.

For complex d, there is no space $\mathbb{R}^d$ in this argument. Define the coefficient denoted by $\left|S^{d-1}\right|$ by the right side of Equation (13.6). For $0 < \operatorname{Re} d < 2\operatorname{Re} s$, the same one-variable substitutions, now with complex exponents, prove

$$\left|S^{d-1}\right| \int_0^\infty r^{d-1}(r^2+m^2)^{-s}\,\mathrm{d}r = \pi^{d/2}\frac{\Gamma(s-d/2)}{\Gamma(s)} m^{d-2s}. \tag{13.7}$$

The endpoint bounds use $\operatorname{Re} d$ and $\operatorname{Re} s$ and prove absolute convergence in precisely this domain. Outside it, the meromorphic right side defines the *dimensional-regularisation prescription* for this integral. This continuation is a prescription, rather than a convergent integral over a space of noninteger dimension.

13.4 The digamma and Laplace's method

The *digamma function* is the logarithmic derivative

$$\psi(s) = \frac{\Gamma'(s)}{\Gamma(s)}.$$

It is meromorphic. The letter ψ here always carries an argument. The wave function denoted by ψ in earlier quantum examples has no role in this section.

On the slit plane $U = \mathbb{C}\setminus(-\infty, 0]$, the quotient $\Gamma'(s)/\Gamma(s)$ is holomorphic and has a primitive by Theorem 9.4, since U is star-shaped about 1. Choose the primitive L with $L(1) = 0$. The derivative of $e^{-L(s)}\Gamma(s)$ vanishes, so this function is constant on the domain and equals 1 at 1. Hence $e^{L(s)} = \Gamma(s)$. On the positive axis $L(s)$ is the real logarithm of the positive function $\Gamma(s)$. We write this branch as $\log\Gamma(s)$. It is not the principal logarithm of the value $\Gamma(s)$.

For functions f and g with g eventually nonzero, the notation $f \sim g$ means $f/g \to 1$. The notation $o(1)$ denotes a quantity tending to zero. We now prove the leading integral estimate used for real Stirling.

Theorem 13.3 (Laplace's method). *Let $I \subset \mathbb{R}$ be an interval and t_0 an interior point. Let $\varphi : I \to \mathbb{R}$ be continuous with unique minimum $\varphi_0 = \varphi(t_0)$. Suppose it is C^2 near t_0 and $\varphi''(t_0) > 0$. Suppose also that outside every sufficiently small neighbourhood of t_0 the difference $\varphi(t) - \varphi_0$ has a strictly positive lower bound. Let the complex amplitude a be measurable, continuous near t_0, and satisfy*

$$\int_I |a(t)|\, e^{-s_0(\varphi(t)-\varphi_0)}\,\mathrm{d}t < \infty \quad \text{for some } s_0 > 0.$$

The integral is absolutely convergent for every real $s \geq s_0$. As the real parameter $s \to +\infty$,

$$\int_I a(t)e^{-s\varphi(t)}\,\mathrm{d}t = \frac{e^{-s\varphi_0}}{\sqrt{s}}\left(a(t_0)\sqrt{\frac{2\pi}{\varphi''(t_0)}} + o(1)\right). \tag{13.8}$$

If $a(t_0) \neq 0$, this is an asymptotic equivalence to its leading term.

Proof. Since $\varphi(t) - \varphi_0 \geq 0$, the weighted integrand for $s \geq s_0$ is bounded in modulus by the integrable one at s_0. Multiplication by the finite constant $e^{-s\varphi_0}$ proves absolute convergence of the stated integral. To prove the limit, take any sequence of real parameters tending to infinity. The following bounds and limits hold along every such sequence.

Choose $\varepsilon > 0$ with $[t_0-\varepsilon, t_0+\varepsilon] \subset I$ and $\varphi(t)-\varphi_0 \geq c(t-t_0)^2$ there for some $c > 0$. Taylor's formula gives this bound, since $\varphi'(t_0) = 0$ and $\varphi''(t_0) > 0$. The amplitude is bounded on this interval. Set $u = \sqrt{s}(t - t_0)$ in the nearby part of the integral, multiply by $\sqrt{s}e^{s\varphi_0}$, and extend the resulting integrand by zero outside $|u| \leq \varepsilon\sqrt{s}$. For each fixed u, Taylor's formula gives the limit

$$a(t_0 + u/\sqrt{s})e^{-s(\varphi(t_0+u/\sqrt{s})-\varphi_0)} \longrightarrow a(t_0)e^{-\varphi''(t_0)u^2/2}.$$

Its modulus is bounded by Ce^{-cu^2}, integrable on $\mathbb{R}$. Dominated convergence and the Gaussian integral give the coefficient in the statement.

On the remaining part of I, let $\varphi(t) - \varphi_0 \geq c_1 > 0$. Its absolute contribution after multiplication by $e^{s\varphi_0}$ is at most

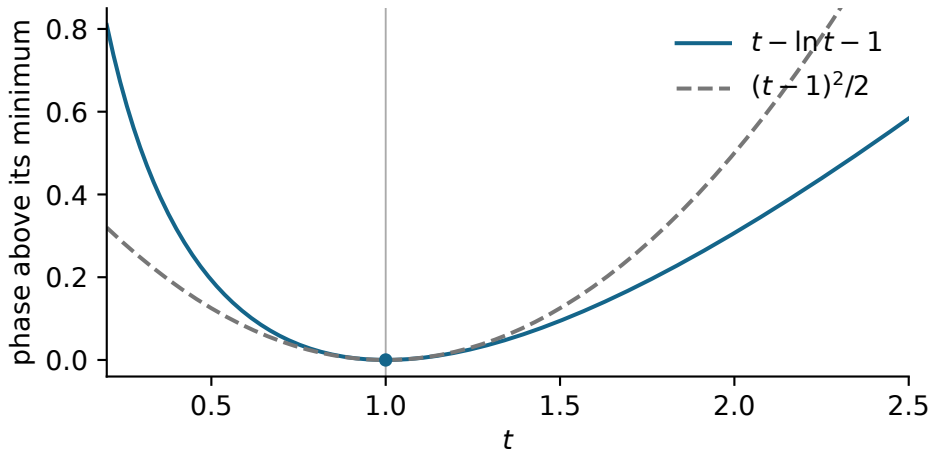
$$e^{-(s-s_0)c_1}\int_I |a(t)|\, e^{-s_0(\varphi(t)-\varphi_0)}\,\mathrm{d}t.$$

Even after multiplication by $\sqrt{s}$ this tends to zero. Adding both parts proves the formula along every such sequence. The sequential criterion for a limit proves the assertion as $s \to +\infty$. ∎

On a compact interval, removing a small open neighbourhood of the unique minimum leaves a compact set. The continuous phase difference is strictly positive there, so it attains a positive minimum. This gives the required gap. On an unbounded interval, neither this compactness argument nor local Taylor expansion controls the distant tail. The gap and the integrable weighted amplitude must be checked separately.

13.5 Real and sectorial Stirling formulas

Apply Laplace's method to $\varphi(t) = t - \ln t$ and $a(t) = 1$ on $(0, \infty)$. Its derivative is $1 - 1/t$. Thus its unique minimum is at $t = 1$, where $\varphi(1) = 1$ and $\varphi''(1) = 1$. The phase tends to infinity at both endpoints, so the required gaps hold. For $s_0 = 1$, the weighted amplitude integral is $e \int_0^\infty te^{-t}\,\mathrm{d}t = e$, which is finite. The quadratic approximation in the proof is shown below.



The phase $t - \ln t - 1$ and its quadratic approximation agree to second order at the unique minimum. Away from that minimum the phase has a positive gap.

Write Euler's integral with integration variable u. Substituting $u = st$ then gives

$$\Gamma(s+1) = s^{s+1} \int_0^\infty e^{-s(t-\ln t)}\,\mathrm{d}t.$$

Hence the proved Laplace estimate gives the following.

Theorem 13.4 (Real Stirling formula). *As $s \to +\infty$ along the positive real axis,*

$$\Gamma(s+1) \sim \sqrt{2\pi s}\, s^s e^{-s}.$$

The equivalence supplies a leading term. It does not by itself bound its relative error by $O(1/s)$.

For sectorial asymptotics we use one cited result. The notation $O(1/|s|)$ uniformly on a set means that the remainder's modulus is at most $C/|s|$ there for all sufficiently large $|s|$, with one constant C for that set.

Theorem 13.5 (Sectorial Stirling formula · *cited*)**.** *On the branch* $\log\Gamma$ *real on the positive axis, for each* $0 < \delta < \pi$*,*

$$\log\Gamma(s) = \left(s - \frac{1}{2}\right)\log s - s + \frac{1}{2}\ln(2\pi) + R(s), \qquad R(s) = O(1/|s|), \tag{13.9}$$

uniformly as $|s| \to \infty$ *in* $|\arg s| \le \pi - \delta$*. The logarithm of* s *is the principal branch on the slit plane.*

This is Beals–Wong, Theorem 2.5.2 and the log-remainder identity (2.5.7), pp. 28–29. The cited estimate is stronger than the real leading equivalence just proved.

Proposition 13.6 (Digamma estimate · *sketched*)**.** *Using the cited sectorial Stirling formula, uniformly on each sector with fixed* $0 < \delta < \pi$*,*

$$\psi(s) = \log s - \frac{1}{2s} + O(1/|s|^2), \qquad |\arg s| \le \pi - \delta, \qquad |s| \to \infty. \tag{13.10}$$

In particular, $\operatorname{Re}\psi(\mathrm{i}y) = \ln y + O(y^{-2})$ *as* $y \to +\infty$*.*

Proof. The argument after the cited input is complete. Use its remainder bound on the larger sector $|\arg s| \le \pi - \delta/2$. For some fixed $c > 0$ depending only on δ, each disk $\overline{D(s, c\,|s|)}$ centred in the smaller sector lies in the larger sector and has $|\zeta| \ge (1-c)\,|s|$ throughout. For example choose $c < 1/2$ with $\arctan(c/(1-c)) < \delta/2$: after rotating by $-\arg s$, every point of the disk has real part at least $(1-c)\,|s|$ and imaginary part of modulus at most $c\,|s|$.

The function R is holomorphic on the slit plane. Its bound on the circle is $C'/|s|$. Cauchy's estimate, Equation (10.3), therefore gives $|R'(s)| \le C'/(c\,|s|^2)$. Differentiating the explicit part of Stirling gives $\log s - 1/(2s)$. Since $(\log\Gamma(s))' = \psi(s)$, this proves the uniform estimate. At $s = \mathrm{i}y$ the real part of $\log(\mathrm{i}y)$ is $\ln y$, and $-1/(2\mathrm{i}y)$ is purely imaginary. Taking real parts proves the final assertion. The tag records the inherited cited input. ∎

Summary.

- Euler's Gamma integral is holomorphic for $\operatorname{Re} s > 0$. Its recurrence continues it with residues $(-1)^n/n!$ at $-n$.
- Beta is a quotient of Gamma functions. Reflection follows from the keyhole integral and the identity theorem. Gamma has no zeros.
- Convergent radial integrals give the stated Gamma quotient. Its continuation defines the dimensional-regularisation prescription.
- At a unique nondegenerate interior minimum, with a positive outer gap and integrable weighted amplitude, Laplace's method gives the Gaussian leading term.
- Real Stirling is proved. Sectorial Stirling is cited, and its holomorphic remainder gives the digamma estimate by Cauchy's bound.

Lecture 14 extends the asymptotic analysis to endpoints and oscillatory integrals.

References

R. Beals and R. Wong, *Special Functions and Orthogonal Polynomials*, Cambridge, 2016, §§2.1–2.2 and 2.5–2.6, pp. 16–22, 26–31. Their Gamma variable z is written s here. Theorem 2.5.2 and identity (2.5.7) are the cited source of the sectorial remainder, with the branch real on the positive axis. V. A. Zorich, *Mathematical Analysis II*, second edition, Springer, 2016, §17.3.3–17.3.5, pp. 438–441 (Beta, sphere area and radial power integrals), and §19.2, pp. 603–610 (Laplace localisation and its leading term). The source's large parameter λ is our s; its maximised phase is our $-\varphi$.